

PROGRAM - 4

TITLE: Integrate Git with Docker for Source-Controlled Application Builds

Project Structure

Program-4

Dockerfile

index.js

package.json

package-lock.json

Step-1: Create a Dockerfile

```
GNU nano 6.2
# stage 1: Build Stage
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .

# stage 2: Runtime stage
FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD ["node","index.js"]
```

Step-2: create package.json by executing the command

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-4$ npm init -y
Wrote to /home/user/labprograms/Program-4/package.json:

{
  "name": "program-4",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

Step-3: create index.js file

```
GNU nano 6.2 index.js
const express=require("express");
const app=express();
const PORT=3000
app.get("/",(req,res)=>{
res.send("Hello from Docker + Git ");
});
app.listen(PORT,(req,res)=>{
console.log("Server running in http://localhost:${PORT}");
});
```

Step-4: install express dependency

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-4$ npm install express
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'express@5.1.0',
npm WARN EBADENGINE   required: { node: '>= 18' },
npm WARN EBADENGINE   current: { node: 'v12.22.9', npm: '8.5.1' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'body-parser@2.2.0',
npm WARN EBADENGINE   required: { node: '>=18' },
npm WARN EBADENGINE   current: { node: 'v12.22.9', npm: '8.5.1' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'content-disposition@1.0.1',
npm WARN EBADENGINE   required: { node: '>=18' },
npm WARN EBADENGINE   current: { node: 'v12.22.9', npm: '8.5.1' }
npm WARN EBADENGINE }

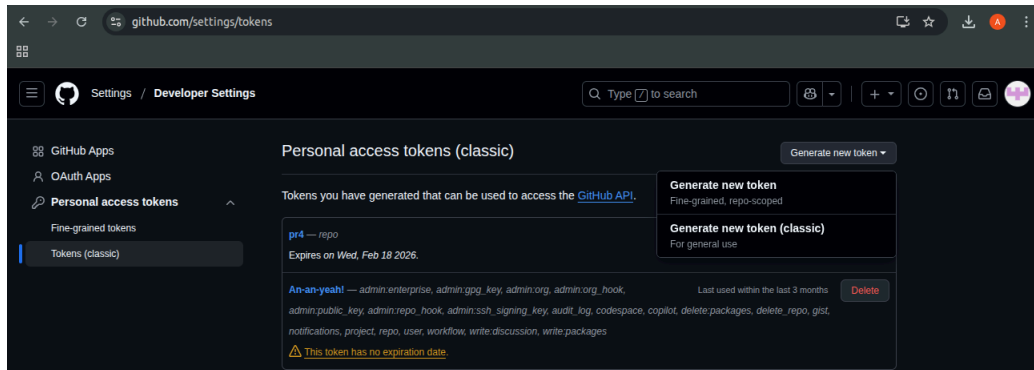
added 68 packages, and audited 69 packages in 2s

18 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

Step 5: Create a PAT and copy the password

Settings—> Developer Settings—> Personal Access Token—> Classic Tokens—> Create token



Step-6: Initialize the git repository locally

- > git init
- > git add .
- > git commit -m "program-4"

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-4$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/user/labprograms/Program-4/.git/
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-4$ git add .
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-4$ git commit -m "program-4"
[master (root-commit) cdbba03] program-4
 4 files changed, 1347 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 index.js
 create mode 100644 package-lock.json
 create mode 100644 package.json
```

Step 7: connect to the repository

- > git remote add https://github.com/Ananyah25/Program-4.git
- > git branch -M origin main

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-4$ git remote add origin https://github.com/Ananyah25/Program-4.git
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-4$ git branch -M main
```

Step 8: push the code

> git push -u origin main

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-4$ git push -u origin main
Username for 'https://github.com': Ananyah25
Password for 'https://Ananyah25@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 8.86 KiB | 1.48 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Ananyah25/Program-4.git
 * [new branch]    main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
```

The screenshot shows the GitHub interface for a repository named 'Program-4' owned by 'Ananyah25'. The repository is public. The main branch is selected, showing 1 branch and 0 tags. The file list includes Dockerfile, index.js, package-lock.json, and package.json, all committed 19 minutes ago. The commit hash is cdbba03 and there is 1 commit.

Program-4 (Public)

main 1 Branch 0 Tags

Go to file Add file Code

Ananyah25 program-4 cdbba03 · 19 minutes ago 1 Commit

Dockerfile	program-4	19 minutes ago
index.js	program-4	19 minutes ago
package-lock.json	program-4	19 minutes ago
package.json	program-4	19 minutes ago